

ECE 537 – Digital Signal Processing

Lecture #1 – Introduction

1.1 Overview of a Digital Signal Processor

- Digital signal processing (DSP) is the analysis and manipulation of signals in digital form. Usually, these signals are physical impressions from the real world, such as sound waves and images. Since there is no way to fully represent the measured data from such a continuous analogue signal in a computer, one has to sample it into discrete values of time and amplitude to get a digital representation. This is done using an analogue to digital converter. Then the processing can take place, which usually involves filtering, measuring or compression. The signal can then be converted back to an analogue signal using a digital to analogue converter.

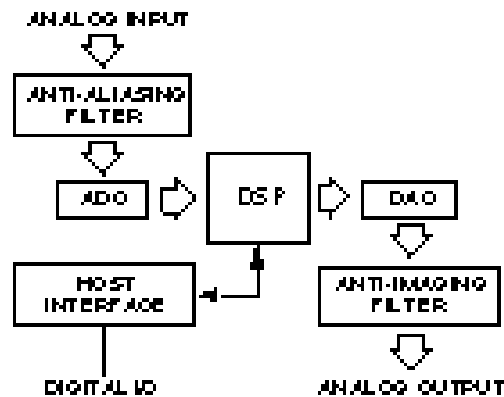


Figure 1.1: Elements of a DSP

- A digital signal processor is a special type of microprocessor which is fabricated on metal oxide semiconductor integrated circuits. DSPs are extensively used in different applications like digital image processing, telecommunications, audio signal processing, speech recognition systems, sonar, radar, etc, and also used in consumer electronics like mobile phones, HDTV (high-definition television) products, disk drives, etc.



Figure 1.2: Digital Signal Processor

The digital signal processor mainly works by using real-world signals like audio, voice, temperature, and video, digitizing them & after that mathematically manipulating them. A DSP performs different mathematical functions very quickly like add, subtract, multiply & divide

- Digital signal processors (DSPs) are digital processors whose hardware, software, and instruction sets are optimized for high-speed numeric processing applications and are essential for processing digital data representing analog signals in real-time.
- The DSPs' high-speed arithmetic and logical hardware is programmed to rapidly execute algorithms modelling a functional transformation.
- Digital signal processors operate on many samples of data per second, require a large memory bandwidth, and perform very intense computations. They can be programmable, as in the case of the Texas Instrument TMS320 series, or dedicated. The principle components of a DSP are a multiplier, adder, shifter, fast registers, and memory.

1.2 Digital Signal Processor Architecture

The architectures of the digital signal processors are;

- Von Neumann Architecture.
- Harvard Architecture.
- Super Harvard Architecture.

- ***Von Neumann Architecture***

Von Neumann's architecture of a digital signal processor mainly includes a single memory & a single bus which are used for data transferring into & out of the CPU (central processing unit). Von Neumann architecture has only one bus which is used for both data transfers and instruction fetches, and therefore data transfers and instruction fetches must be scheduled - they cannot be performed at the same time.

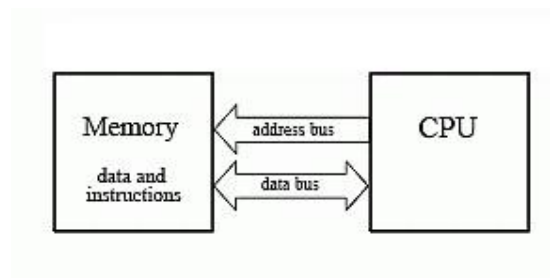


Figure 1.3: Von Neumann Architecture

This type of architecture is quite suitable when serial operation of all of the necessary tasks is satisfactory. At present, most computers use Von Neumann architecture.

- **Harvard Architecture**

The name Harvard Architecture as shown in Figure 4.1, it includes two separate memories for both the data & program instructions including separate buses for each. When the buses work independently, then data & program instructions can be fetched together to improve the speed over the single bus. At present, this dual bus architecture is used by DSPs including TI TMS320C10.

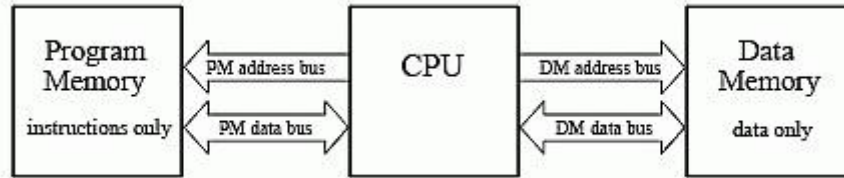


Figure 1.4: Harvard Architecture

- **Super Harvard Architecture**

The super Harvard architecture of DSP is shown Figure 1.5. This name was coined through Analog Devices to explain the internal function of their new ADSP-211xx & ADSP-2106x families of DSPs which are called SHARC DSP which is a reduction of the longer term of Super Harvard Architecture.

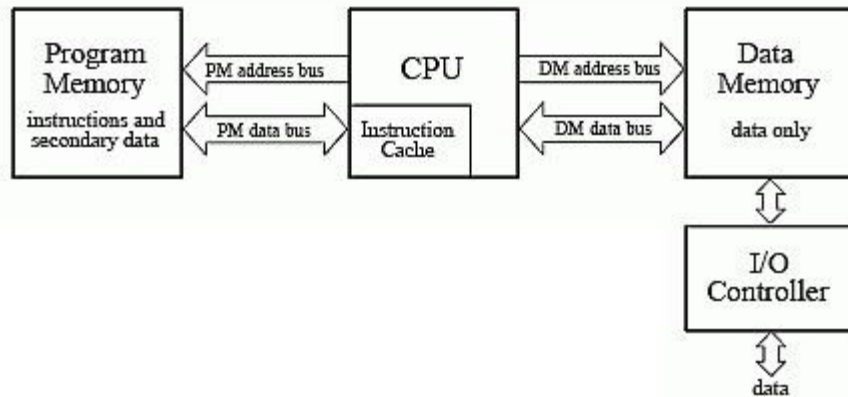


Figure 1.5: Super Harvard Architecture

This architecture was implemented by including some features to increase the throughput. While the super Harvard architecture digital signal processors are optimized in several methods, two areas are significant enough to be included like an instruction cache & an I/O controller.

1.3 Features of digital signal processors

The features of digital signal processors include the following.

- Digital signal processors are mainly designed for supporting repetitive and numerically intensive tasks.
- Most digital signal processors include a powerful data path and also the capacity to move large amounts of data to memory quickly.
- These processors provide unique instruction sets to develop the efficiency of hardware.

- The two significant features of digital signal processors are; the data path including multiple-access memory architectures & fast multiply-accumulate units.
- Pipelining is frequently used to enhance the processor's performance. So at present, most processors use pipelining but in the development of enhancing performance, pipelining will make programming very hard.

1.4 Characteristics of Digital Signal Processors

Digital Signal Processors (DSPs) are microprocessors with the following characteristics:

- a) Real-time digital signal processing capabilities: DSPs typically have to process data in real time, i.e., the correctness of the operation depends heavily on the time when the data processing is completed.
- b) High throughput: DSPs can sustain processing of high-speed streaming data, such as audio and multimedia data processing.
- c) Deterministic operation: The execution time of DSP programs can be foreseen accurately, thus guaranteeing a repeatable, desired performance.
- d) Re-programmability by software: Different system behaviour might be obtained by re-coding the algorithm executed by the DSP instead of by hardware modifications.

1.5 Types of Digital Signal Processor

Digital signal processors are available in two types: fixed-point processors and floating-point processors.

- ***Fixed Point Digital Signal Processor***

In a fixed-point digital signal processor, every number can be specified through a minimum of 16 bits, even though a different length can be utilized. The number can be represented with different patterns.

The fixed-point means that the fractional point position can be assumed to be fixed and to be identical for the operands as well as the operation result.

Fixed point processors are used in different flexible embedded applications because it uses low power and less cost. Fixed-point digital signal processors include: TI's TM320C54x, ADI DSP BF53X, TM320C55x, TM320C64x, TM320C62x and Motorola MSC810x.

- ***Floating Point Digital Signal Processor***

Floating-point digital signal processors mainly use a minimum of 32 bits to store every value. The distinct feature of floating-point DSP is that the signified numbers are not spaced uniformly. Floating-point digital signal processors can simply process the fixed-point numbers, a requirement to implement counters & signals which are received from the analog to digital converter and transmitted to the digital to analog converter.



Figure 1.6: Floating Point DSP

As compared to fixed-point DSPs, the programs of floating-point DSPs are simple, however, they are normally very expensive and power consumption is also more. Examples of floating-point DSPs are TI's TMS320c67x and ADI ADSP 2116x/2126x.

1.6 Applications of DSP

DSPs appeared on the market in the early 1980s. Over the last 35 years they have been the key enabling technology for many electronics products in fields such as communication systems, multimedia, automotive, instrumentation and military.

DSP applications mainly include processing of audio & speech, radar, sonar & other sensor array processing, statistical signal processing, spectral density estimation, data compression, digital image processing, audio coding, video coding, image compression, signal processing for control systems, telecommunications, seismology, biomedical engineering, etc.

Thus, a digital signal processor is a specialized microprocessor. The main function of this processor is to measure, and compress and filter analog signals. Usually, digital signal processors have better power efficiency, so they are mostly used in portable devices like mobile phones due to power utilization constraints. These processors frequently use special memory architectures to fetch multiple data otherwise instructions simultaneously.

1.7 Programming DSP

DSP algorithms are run on various platforms, often on standard computers, but also frequently on specialized microprocessors, digital signal processors (DSPs). When speed is a major concern, field-programmable gate arrays (FPGA) and application-specific integrated circuits (ASIC) might be used.

Most digital signal processing code is written in low level C, highly optimized for the specific target platform, due to performance demands. High performance is usually crucial, since many applications, especially in telecommunications, have real-time demands. Writing the algorithms in C usually works well, but when an application is to be moved to a different target platform, it might be necessary to rewrite the entire code because of all the optimizations. This is time consuming and error prone, which makes it a very expensive task. Also, the C code is not very intuitive to read because of its machine-oriented nature. Heavy use of intrinsic instructions and other adaptations for specific hardware makes it hard to read and maintain. The demands on DSP

software will probably increase rapidly over the next few years. A reason for this is the never ending pursuit of higher bit-rates in mobile communications. Instead, an intuitive and maintainable high-level language would be preferred for writing the DSP algorithms. This language could then be compiled into target code for different platforms, preferably with multi-core support, and without sacrificing performance. However, most DSP targets today only have C compilers, and it is standard to write the software in C. Efficient code generation from a high-level language to C code is thus a subject which seems to be of increasing interest. Below, a few examples of such products are briefly discussed.

- * Feldspar is a high-level functional programming language targeting DSP applications. Since it is a functional language, programs in Feldspar usually follow a data-flow programming style, meaning that computations are described as networks of data structure transformations. The language is a work in progress with objectives of increasing maintainability and portability without sacrificing performance. Currently, there is an ISO C99 back-end available, as well as limited support for the TIC64x family, but the intention is to provide back-ends for many DSPs in the future.
- * MATLAB is a numerical computing environment and high-level programming language focused around vector and matrix manipulations. It features a large amount of built-in functions for many kinds of mathematical calculations, and also has built-in visualization functionalities for easy data analysis. MATLAB Coder is a tool for generating C code from MATLAB code, which supports code generation for many different target platforms. A subset of the MATLAB language is supported for C code generation, including most functionalities needed for algorithm design. This subset together with a compiler for C code generation was referred to as Embedded MATLAB.

1.8 Difference between Digital Signal Processor and Conventional Microprocessor

Digital Signal Processor	Microprocessor
It is a specialized microprocessor chip	It is a computer processor
DSPs are extensively used in telecommunications, audio signal processing, digital image processing, etc	Microprocessors are used in PCs for text editing, computation, multimedia display & communication over the Internet.
In DSP, instruction can be simply executed in a single CLK cycle.	The microprocessor uses several clock cycles for one instruction execution.
Parallel execution can be achievable	Sequential execution is possible.
DSP is suitable for the operation of array processing.	It is suitable for general-purpose processing.
Addressing modes used in this processor are direct & indirect.	Addressing modes used in microprocessors are direct, immediate, register indirect, indirect register, etc.
Address generation can be possible by combining program sequencers & DAGs.	The program counter or PC can be incremented to produce an address sequentially.
It includes three separate computational units: MAC, ALU & Sifter.	It includes simply the main unit like ALU.
It is a specialized microprocessor chip	It is a computer processor
The program flow can be controlled by an instruction register & program sequencer.	Program counter can control the execution flow.
It includes separate data & program memories.	It doesn't have separate memories.

In DSP, several operands are fetched at once.	In a microprocessor, the operand can be fetched serially.
In DSP, address & data bus are multiplexed	In a microprocessor, address & data bus are not multiplexed.

1.9 Advantages and Disadvantages of Digital Signal Processing

The advantages of a digital signal processing over analog include the following:

- Overall noise is less.
- Possibility of error detection & correction.
- Simple data storage.
- Digital signals are simple to encrypt.
- Transmission of more data is possible.
- In digital processing systems, modifying a few commands otherwise change a few code lines is easy to modify.
- DSP systems work through a broader range of frequencies.
- In a digital system, the DSP can be cascaded without any loading issues.
- The operations of DSP can be simply changed by altering the program within a digital programmable system.
- By using the DSP method, a complicated signal processing algorithm can be simply implemented.
- DSPs are lightweight & more compact.
- DSP systems are upgradeable because they are controlled by software.

Disadvantages

The disadvantages of a digital signal processing include the following:

- Digital communications need high bandwidth to transmit the data as compared to analog.
- Most of the digital signal processors are expensive.
- The DSP system complexity will be increased due to the usage of additional components.
- Digital signal processor uses several transistors which consume more power as compared to analog signal processors.
- The hardware architecture & software instructions of each DSP are different so required highly skilled engineers to program the device.

Homework

Q1

- Briefly outline four (4) of the main advantages and one (1) disadvantage of digital signal processing (DSP) as opposed to analogue signal processing.
- Digital signal processors are widely used in embedded system design, describe the design and architecture of a conventional digital signal processor.
- Embedded systems designs can be implemented either using single chip microcontrollers or digital signal processor chips (DSP chips). List the differences between using microcontroller and using DSP chip.

- Q2. Signal processing has always played a critical role in the development of telecommunication systems. In the current standards of terrestrial mobile cellular communication many solutions for base stations or mobile stations have been implemented using, Application Specific Integrated Circuits (ASICs), and Digital Signal Processors (DSPs); using labeled block diagrams describe typical applications of DSPs in GSM cellular radio communication.
- Q3. A common example of a digital signal processing application is the removal of the noise component from a signal corrupted by additive noise.

Let $s[n]$ be the signal corrupted by a random noise $d[n]$ resulting in the noisy signal:

$$x[n] = s[n] + d[n] \quad (1)$$

The objective is to operate on $x[n]$ to generate a signal $y[n]$ which is a reasonable approximation to $s[n]$. To this end, a simple approach is to generate an output sample by averaging a number of input samples around the sample at instant n . A three-point Moving Average filter algorithm is described by the following input/output equation:

$$y[n] = 1/3 (x[n-1] + x[n] + x[n+1]). \quad (2)$$

- Write a MATLAB script to implement the above algorithm:
- Generate and show all pertinent signals.
- What is the form of the uncorrupted signal $s[n]$?
- What is the form of the additive noise $d[n]$?